

AI531C — Week 5 Pre-Class Activity (PCM-5: Pick the Right Filter)

Prepares: Lecture 11 (Spatial denoising, Tue 1 Sep) · Lectures 12–13 (Wiener restoration; Geometric transforms), Wed 2 Sep · Tutorial T5 (Tue 1 Sep).

1 · Topic

Matching the filter to the noise: mean, median, Gaussian and bilateral — and why one filter's triumph is another noise's disaster.

2 · Why you are preparing this

Lecture 11 runs a measured tournament (PSNR per filter per noise). Arriving with a prediction of the bracket lets the class argue about WHY the median demolishes salt-and-pepper while the mean smears it.

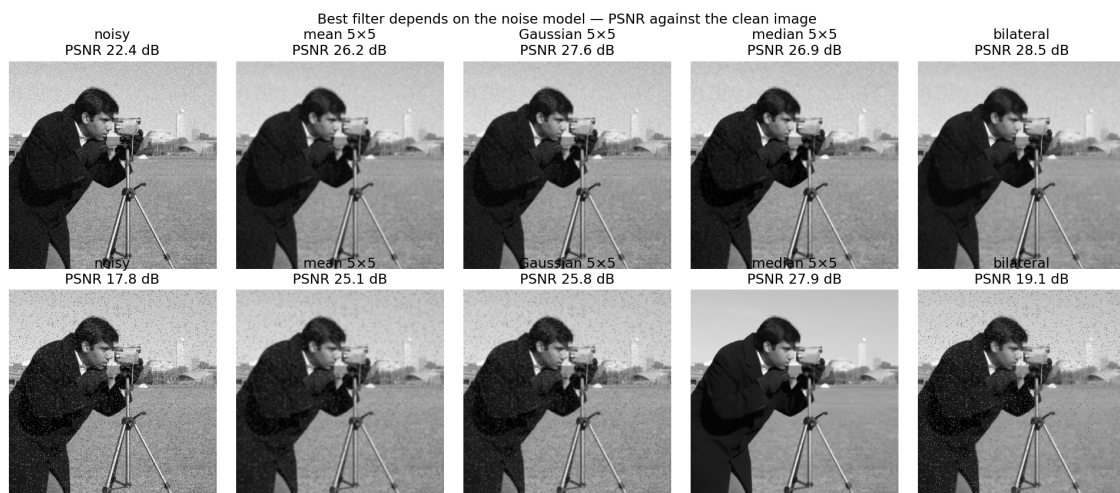
3 · Read (12–15 min)

From the Comprehensive Self-Contained Lecture Notes:

- Lecture 11 chapter, the sections on the four spatial filters (about 3 pages);
- Lecture 12 chapter, ONLY the first page (the inverse-filter idea and its danger).

Optional depth: GW4 Ch. 5, Secs. 5.3 (order-statistics filters).

4 · Interpret one figure (4 min)



Source: course figure, Lecture 11 (filter × noise tournament grid).

In two sentences: locate the best and worst cells of the grid, and state what property of the median filter explains BOTH.

5 · One small calculation (5 min)

A 3×3 mean filter averages 9 i.i.d. noise samples of $\sigma = 20$. Compute the output noise σ . What did the same operation do to a 1-pixel-wide edge?

6 · Predict (4 min)

Predict the tournament: rank {mean, median, Gaussian} on salt-and-pepper noise BEFORE class. One line of justification for your #1.

7 · Inspect this fragment (3 min)

```
den = cv2.medianBlur(noisy, 5)
```

```
psnr = 10*np.log10(255**2 / np.mean((img-den)**2))  
# median of an ODD count of values – why odd?
```

Question: why do median filters use odd window sizes? What would a 4×4 median have to invent?

8 · Bring one question to class

One question about denoising — or one case where you suspect NO spatial filter can win (bring it; Lecture 12's frequency view may answer it).

Submit: items 4-6 — half a page maximum, before Tue 1 Sep, 9:00 AM.

Expected time: 30–35 minutes. Counts toward the weekly preparation portfolio (best 10 of PCM-2...PCM-14, completion-based micro-credit per the lesson plan). The short per-lecture primers in the course folder are optional warm-ups; only THIS card is submitted.